

Document number	PLS-SDS-MP-HIV-001
Revision	Reconciled Draft v0.2
Issue date	09 October 2026
Product format	Lyophilized PCR tube-strip microchips; control presentation subject to final approved bill of materials
Pack sizes	48 and 96 tests where supported by a product-specific supplier IFU; other claims require confirmation
Evidence status	Product-specific supplier IFU available.

Important: This is an evidence-reconciled working draft. It is not approved for external release. Final classification requires the complete quantitative formulation, raw-material SDS/CoAs, stability evidence, verified company/emergency contacts and Quality authorization.

1. Identification

Product identifier	Mugen-Plex HIV-1 — Nucleic Acid Detection Kit (Lyophilized Fluorescent PCR Method)
Recommended use	Qualitative laboratory detection of human immunodeficiency virus type 1 (HIV-1) RNA in serum by one-step reverse-transcription real-time fluorescent PCR; for trained laboratory personnel.
Restrictions on use	For research use only unless and until otherwise authorized. Not for self-testing. Not to be used alone to confirm or exclude infection.
Brand owner / local contact	Precision Life Sciences (Pvt.) Ltd. [legal name to be verified] Office #09, Police Colony, Nasir Bagh Road, Peshawar, Khyber Pakhtunkhwa, Pakistan [address from uploaded SDS draft] info@precisionlifesciences.com.pk +92 301 8182903 [contact from registration inserts] PrecisionLifeSciences.com.pk
Original reagent manufacturer	Shanghai Janzy Biotechnology Co., Ltd. Supplier contact stated in IFU: Ms. Qiao, 18105338426
Emergency information	No verified dedicated emergency number was supplied. For exposure, contact local medical services and provide this SDS; for fire/spill, follow the facility emergency procedure. Insert a verified staffed number before controlled external release.

2. Hazard identification

GHS classification	No GHS hazard classification is identified from the ingredient information currently available and the small lyophilized quantities described. Final classification remains dependent on complete quantitative formulation and raw-material supplier SDS review.
Signal word / pictograms	None assigned on the available information.
Other hazards	Reconstituted or used tubes may contain human specimen nucleic acid and must be handled as potentially biohazardous. Lyophilized material and nucleic-acid controls are not identified as infectious virus. Avoid aerosol generation and contamination.

3. Composition / information on ingredients

Detection microchip	Primers, dye-labelled hydrolysis probes, reverse transcriptase, Taq DNA polymerase, Tris, potassium chloride, magnesium sulfate and dNTPs. Quantitative concentrations and a complete excipient/preservative declaration were not supplied.
Controls	Supplier IFU describes positive, negative and exogenous internal controls as plasmid material; no live virus or human-derived control material is identified.

Hazardous ingredients	None identified at reportable concentration from available information. Obtain and retain supplier SDS/CoA/origin declarations for all raw materials before final controlled approval.
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4. First-aid measures

Inhalation	Move to fresh air if dust/aerosol is inhaled. Seek medical advice if symptoms persist.
Skin contact	Remove contaminated clothing. Wash skin with soap and water.
Eye contact	Rinse cautiously with clean water for at least 15 minutes; remove contact lenses if easy to do. Seek medical advice for persistent irritation.
Ingestion	Rinse mouth. Do not induce vomiting. Seek medical advice and show this SDS.
Notes to physician	Treat symptomatically. No specific antidote is identified.

5. Fire-fighting measures

Suitable media	Use water spray, dry chemical, carbon dioxide or foam appropriate to the surrounding fire.
Specific hazards	The supplied kit is not identified as flammable. Thermal decomposition of packaging or reagents may produce irritating fumes.
Protective equipment	Firefighters should use standard protective equipment and self-contained breathing apparatus where smoke or decomposition products are present.

6. Accidental release measures

Personal precautions	Restrict access; wear laboratory coat, gloves and eye protection.
Cleanup	Avoid raising dust or generating aerosols. For unopened/dry material, collect carefully into a closable container and wipe the area with water followed by an appropriate laboratory disinfectant. For used or specimen-contaminated material, follow the facility biohazard spill procedure.
Environmental precautions	Prevent uncontrolled release to drains or the environment.

7. Handling and storage

Handling	For trained laboratory personnel. Keep pre-amplification, specimen-processing and amplification areas separated. Do not mix components from different lots. Prevent moisture, aerosol and nucleic-acid contamination.
Unopened storage	Product-specific supplier IFU: store and transport dry at room temperature; 24 months unopened.
After opening	When resealed, up to 3 days; for longer storage after opening, freeze at -20°C .
Reconstituted controls	For products with a reviewed IFU: store at -25 to -10°C in aliquots and avoid repeated freeze-thaw cycles. For source-limited products, this claim remains unverified.
Shelf life	24 months unopened only where stated by a product-specific supplier IFU; every final label claim still requires approved stability evidence.

8. Exposure controls / personal protection

Occupational limits	No specific occupational exposure limits identified for the disclosed mixture.
Engineering controls	Use normal molecular-laboratory ventilation and physical separation of clean, extraction and amplification areas. Use a biological safety cabinet when handling potentially infectious human specimens.
PPE	Laboratory coat, disposable gloves and safety glasses. Add face protection and other PPE according to specimen risk assessment and spill procedure.
Hygiene	Do not eat, drink or apply cosmetics in the laboratory. Wash hands after handling and before leaving the laboratory.

9. Physical and chemical properties

Appearance	Lyophilized microsphere/reagent in PCR tube-strip microchips. Final bill of materials and control presentation must be fixed in the approved device master record.
Odour	Not determined / expected to be minimal.
pH, melting point, boiling point, vapour pressure, density, solubility, flash point	Not determined for the mixture.

10. Stability and reactivity

Reactivity Stability	No hazardous reactivity identified under recommended handling. Expected to be stable when kept dry and stored as specified in the product-specific supplier IFU.
Conditions to avoid	Moisture, excessive heat, direct sunlight, contamination, use after expiry and incompatible lot mixing.
Incompatible materials	Strong oxidizing agents and conditions that denature enzymes or damage nucleic acids.
Hazardous decomposition	May emit irritating fumes if packaging or organic components burn.

11. Toxicological information

Likely routes Acute/chronic effects	Eye/skin contact, inhalation of dust/aerosol, accidental ingestion. No mixture-specific toxicology studies or quantitative data were provided. No hazardous component is identified at a reportable concentration from available information.
Sensitization/carcinogenicity/mutagenicity/reproductive toxicity	No data available for the finished mixture. Enzyme preparations may cause sensitization in susceptible individuals after repeated occupational exposure.

12. Ecological information

Ecotoxicity, persistence, bioaccumulation and mobility Precaution	No finished-mixture data available. Avoid uncontrolled release. Dispose of used assay materials through the approved laboratory-waste stream.
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13. Disposal considerations

Unused product	Dispose according to institutional chemical/laboratory-waste procedures and applicable Pakistani requirements.
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Used/specimen-contact materials	Treat as potentially infectious laboratory waste. Segregate, decontaminate/autoclave where required and hand over through the licensed healthcare-waste route. Do not dispose of specimen-contact material as general waste.
Packaging	Uncontaminated packaging may follow approved recycling/general-waste procedures; contaminated packaging follows the biohazard stream.

14. Transport information

UN number / proper shipping name / class / packing group Transport status	Not assigned from the available disclosed composition. The unused finished kit is not identified as dangerous goods on the available information. Final shipper classification must be confirmed after complete formulation and supplier SDS review and must account for any dry ice used for reconstituted controls.
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15. Regulatory information

Pakistan	Product registration is governed by the DRAP Act, 2012 and Medical Devices Rules, 2017, as applicable. This SDS does not replace product registration, approved labelling, biosafety procedures or raw-material SDS.
Workplace classification	Final GHS classification and label elements require confirmation against the complete quantitative formulation and supplier component SDS.

16. Other information

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Evidence status	Product-specific supplier IFU available.
Sources	HIV-1 English IFU, revised 15 August 2026; uploaded SDS draft v0.1; KMU Quality Manual; KMU Waste Segregation Protocol; DRAP Medical Devices Rules, 2017.
Approval limitation	DRAFT — NOT FOR EXTERNAL RELEASE. Technical/Quality authorization is required. Complete raw-material SDS/CoAs, quantitative formulation and the product-specific open items must be closed for final hazard classification.
Open approval items	Resolve serum versus plasma and dry room-temperature versus 2–8 °C/-20 °C claims; confirm quantitative formulation, control identities, final kit configuration, REF and emergency contact.

Authorization

Prepared by	_____	Date: _____
Technical review	_____	Date: _____
Quality approval	_____	Date: _____
Authorized release	_____	Date: _____